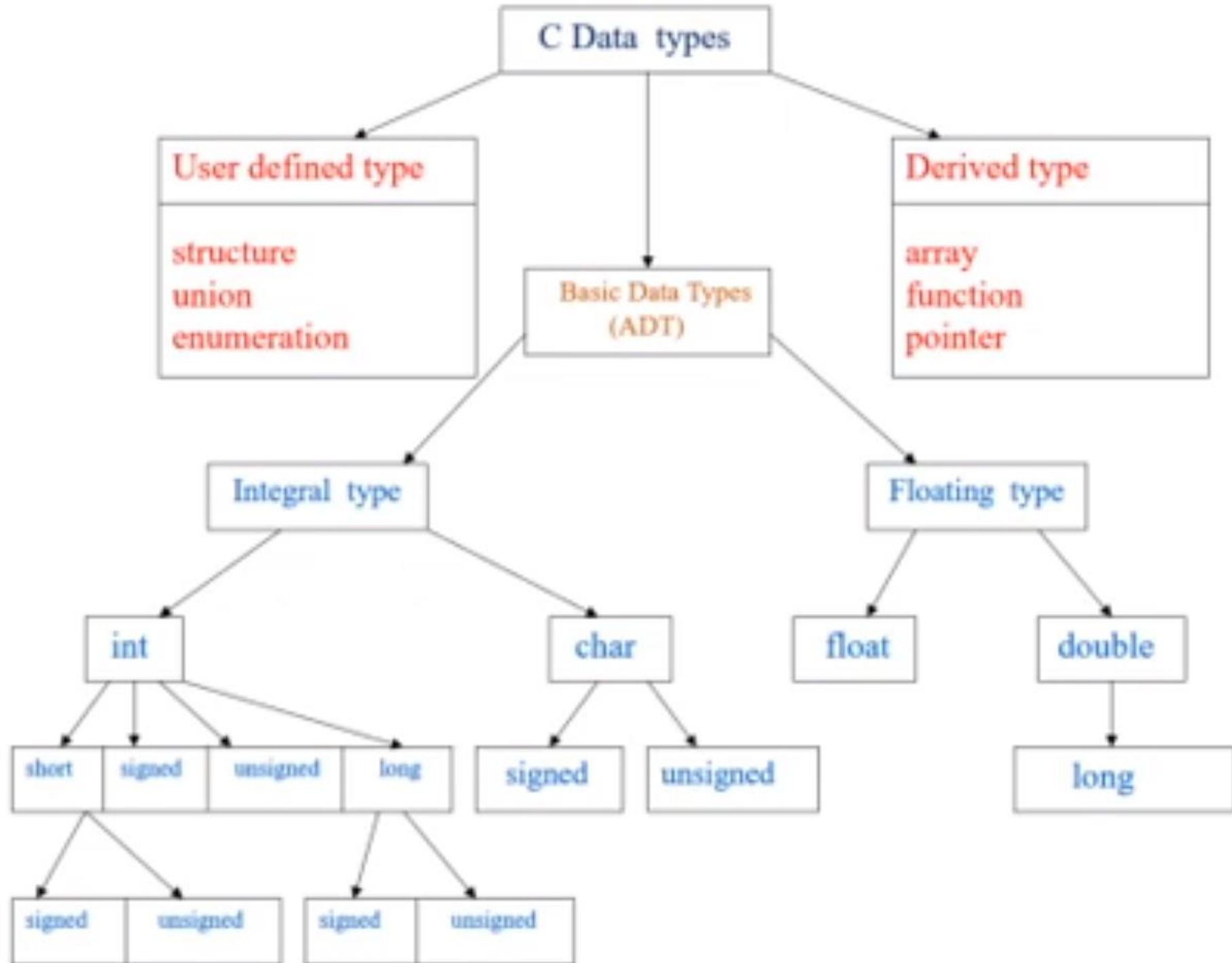




Structure

- It is a user defined data type.
- It creates a data type that can be used to group items of possibly different types into a single type.

Why?

- To group logically related data items of different data types into a single unit
- To access records in database applications

Syntax:

```
struct udsn  
{  
    datatype1 vn1;  
    datatype2 vn2;  
    ...  
};
```

struct – is a keyword
udsn – user defined structure name
datatype1, datatype2 – valid data types in C such as int, float etc

Example - Structure

```
struct Student
{
    char name[25];
    int regno;
    float average;
};
```



Ex.

```
struct date
{
    int day, month, year;
};
```

- Individual components of a struct are called Members fields
- Members can be of different types (simple, array or struct).
- Members can be accessed by struct variable with dot operator
- Memory not allocated for struct.

Structure Variable

- A structure variable can either be declared with structure declaration or as a separate declaration like basic types.
- **Memory allocated for structure variables.**

Ex1.

```
struct student  
{  
    char name[25];  
    int regno;  
    float avg;  
} s1;
```

// s1 known as structure variable and memory
// allocated $(25+2+4)= 31$ Bytes

s1	name	regno	avg
→	25	2	4

Ex2.

```
struct student  
{  
    char name[25];  
    int regno;  
    float avg;  
};  
void main()  
{  
    struct student s1;  
}
```

Access the members

- struct members are accessed by struct variables with dot(.) operator.
- Syntax: **structvariable.member;**

```
#include<stdio.h>
```

```
struct student
```

```
{    char name[25];  
    int regno;  
    float avg;
```

```
};
```

```
void main()
```

```
{  
    printf("Enter name, regno and avg");  
    scanf("%s%d%f",&s1.name,&s1.regno,&s1.avg);  
    printf("Name is = %s\n",s1.name);  
    printf("Register Number is = %d\n",s1.regno);  
    printf("Average Marks = %f\n",s1.avg);  
}
```

Enter name, regno and avg

RajaRam

3456

78.87

Name is = RajaRam

Register Number is = 3456

Average Marks = 78.870003

Size of structure

```
#include<stdio.h>
struct student
{
    char name[25];
    int regno;
    float avg;
};
void main()
{
    struct student s1;
    printf("Enter name, regno and avg");
    scanf("%s%d%f",&s1.name,&s1.regno,&s1.avg);
    printf("Name is = %s\n",s1.name);
    printf("Register Number is = %d\n",s1.regno);
    printf("Average Marks = %f\n",s1.avg);
    printf("Size is = %d",sizeof(s1));
}
```

Enter name, regno and avg

RajaRam

3456

78.87

Name is = RajaRam

Register Number is = 3456

Average Marks = 78.870003

Size is = 31

Initialization in main

```
#include<stdio.h>
struct student
{
    char name[25];
    int regno;
    float avg;
};
```

Name	RegNo	Average
sankar	12345	75.750000

```
void main()
{
    struct student s1={"sankar",12345,75.75};
    printf("Name\t\tRegNo\t\tAverage\n");
    printf("%s\t\t%d\t\t%f\n",s1.name,s1.regno,s1.avg);
}
```

Initialization in struct - Error

```
#include<stdio.h>
struct student
{
    char name[25];
    int regno=12345;
    float avg;
};
```

Error

```
void main()
{
    struct student s1={"sankar",12345,75.75};
    printf("Name\t\tRegNo\t\tAverage\n");
    printf("%s\t\t%d\t\t%f\n",s1.name,s1.regno,s1.avg);
}
```

Array variable in struct

```
void main()
{
    struct student s1;
    int i;
    printf("Enter name, regno");
    scanf("%s%d",&s1.name,&s1.regno);
    printf("Enter three marks\n");
    s1.total=0;
    for(i=0;i<3;i++)
    {
        scanf("%d",&s1.marks[i]);
        s1.total=s1.total+s1.marks[i];
    }
    s1.avg=s1.total/3.0;
    printf("Name is = %s\n",s1.name);
    printf("Register Number is = %d\n",s1.regno);
    printf("Average Marks = %f\n",s1.avg);
    printf("Size is = %d",sizeof(s1));
}
```

```
#include<stdio.h>
struct student
{
    char name[25];
    int regno, marks[3],total;
    float avg;
};
```

Enter name, regno
sankar
12345
Enter three marks
88
77
99
Name is = sankar
Register Number is = 12345
Average Marks = 88.000000
Size is = 39

Creation of more than one variables

```
#include<stdio.h>
```

```
struct student
```

```
{  
    char name[25];  
    int regno;  
    float avg;  
};
```

Size is = 31

Enter name, regno and avg

ganesh

3456

66.567

Name	RegNo	Average
sankar	12345	75.750000
ganesh	3456	66.567001

```
void main()
```

```
{  
    struct student s1={"sankar",12345,75.75};  
    struct student s2;  
    printf("Size is = %d\n",sizeof(s2));  
    printf("Enter name, regno and avg\n");  
    scanf("%s%d%f",&s2.name,&s2.regno,&s2.avg);  
    printf("Name\t\tRegNo\t\tAverage\n");  
    printf("%s\t\t%d\t\t%f\n",s1.name,s1.regno,s1.avg);  
    printf("%s\t\t%d\t\t%f\n",s2.name,s2.regno,s2.avg);  
}
```

Nested Structures

- Structure is defined inside another structure known as nested structures.
- **Types**
 - By Separate structure
 - **By Embedded structure**
- Accessing the members of outer structure is as follows:
outerstructvariable.member;
- Accessing the members of inner structure is as follows:
outerstructvariable.innerstructvariable.member;

Separate Structure

- It contains minimum of two structures and extended up to any level.
- One structure should be used inside other structure as a member.

- Ex.
struct date
{
 int day, month, year;
};
struct student
{
 char name[25];
 int regno;
 float avg;
 struct date dob;
};

- Here dob is a member of structure student.

Separate Structure - Program

```
#include<stdio.h>
struct date
{
    int day, month, year;
};
struct student
{
    char name[25];
    int regno;
    float avg;
    struct date dob;
};
```

```
void main()
{
    struct student s1;
    printf("Size is = %d\n",sizeof(s1));
    printf("Enter name, regno and avg\n");
    scanf("%s%d%f",&s1.name,&s1.regno,&s1.avg);
    printf("Enter day,month and year for dob\n");
    scanf("%d%d%d",&s1.dob.day,&s1.dob.month,&s1.dob.year);
    printf("Name\t\tRegNo\t\tAverage\t\tDOB\n");
    printf("%s\t\t%d\t\t%f\t\t",s1.name,s1.regno,s1.avg );
    printf("%d-%d-%d\n",s1.dob.day,s1.dob.month,s1.dob.year );
}
```

Separate Structure - Program

Size is = 37

Enter name, regno and avg

sankarganesh

12345

68.746

Enter day,month and year for dob

23

04

1976

Name

RegNo

Average

DOB

sankarganesh

12345

68.746002

23-4-1976

Separate Structure - Initialization

```
#include<stdio.h>
```

```
struct date
```

```
{  
    int day, month, year;  
};
```

```
struct student
```

```
{  
    char name[25];  
    int regno;  
    float avg;  
    struct date dob;  
};
```

```
void main()
```

```
{  
    struct student s1={"Raja",11223,65.89,{12,12,1974}};  
    printf("Size is = %d\n",sizeof(s1));  
    printf("Name\t\tRegNo\t\tAverage\t\tDOB\n");  
    printf("%s\t\t%d\t\t%f\t\t",s1.name,s1.regno,s1.avg );  
    printf("%d-%d-%d\n",s1.dob.day,s1.dob.month,s1.dob.year );  
}
```

Size is = 37

Name	RegNo	Average	DOB
Raja	11223	65.894002	12-12-1974

Embedded Structure

- It contains minimum of two structures and extended up to any level.
- One structure should be declared inside other structure.
- Ex.

```
struct student
{
    char name[25];
    int regno;
    float avg;
    struct date
    {
        int day, month, year;
    };
};
```

Embedded Structure - Program

```
#include<stdio.h>
struct student
{
    char name[25];
    int regno;
    float avg;
    struct date
    {
        int day, month, year;
    }dob;
};
```

```
void main()
{
    struct student s1;
    printf("Enter name, regno and avg\n");
    scanf("%s%d%f",&s1.name,&s1.regno,&s1.avg);
    printf("Enter day,month and year for dob\n");
    scanf("%d%d%d",&s1.dob.day,&s1.dob.month,&s1.dob.year);
    printf("Name\t\tRegNo\t\tAverage\t\tDOB\n");
    printf("%s\t\t%d\t\t%f\t\t",s1.name,s1.regno,s1.avg );
    printf("%d-%d-%d\n",s1.dob.day,s1.dob.month,s1.dob.year );
}
```

Embedded Structure - Program

Enter name, regno and avg

sachin

10101

88.9746

Enter day,month and year for dob

23

04

1974

Name

RegNo

Average

DOB

sachin

10101

88.974602

23-4-1974

Embedded Structure - Initialization

```
#include<stdio.h>
struct student
{
    char name[25];
    int regno;
    float avg;
    struct date
    {
        int day, month, year;
    }dob;
};
```

```
void main()
{
    struct student s1={"Raja",11223,65.89,{12,12,1974}};
    printf("Size is = %d\n",sizeof(s1));
    printf("Name\t\tRegNo\t\tAverage\t\tDOB\n");
    printf("%s\t\t%d\t\t%f\t\t",s1.name,s1.regno,s1.avg );
    printf("%d-%d-%d\n",s1.dob.day,s1.dob.month,s1.dob.year );
}
```

Size is = 37

Name	RegNo	Average	DOB
Raja	11223	65.894002	12-12-1974

Passing Structure to function

- Like variables, a structure can also be passed to a function.
- May pass the structure members into the function or pass the structure variable at once.
- Program – Passing struct members

```
#include<stdio.h>
struct student
{
    char name[25];
    int regno;
    float avg;
};
```

```
void display(char n[], int rno, float a)
{
    printf("Name\t\tRegNo\t\tAverage\n");
    printf("%s\t\t%d\t\t%f\n",n,rno,a);
}
```

```
void main()
{
    struct student s1;
    printf("Enter name, regno and avg\n");
    scanf("%s%d%f",&s1.name,&s1.regno,&s1.avg);
    display(s1.name,s1.regno,s1.avg);
}
```

Enter name, regno and avg

sankarganesh

12345

68.746

Name

RegNo

Average

sankarganesh

12345

68.746002

Passing struct variable to function

```
#include<stdio.h>
struct student
{
    char name[25];
    int regno;
    float avg;
};
```

```
void display(struct student s1)
{
    printf("Name\t\tRegNo\t\tAverage\n");
    printf("%s\t\t%d\t\t%f\n",s1.name,s1.regno,s1.avg);
}
```

```
void main()
{
    struct student s;
    printf("Enter name, regno and avg\n");
    scanf("%s%d%f",&s.name,&s.regno,&s.avg);
    display(s);
}
```

Enter name, regno and avg

sankarganesh

12345

68.746

Name

RegNo

Average

sankarganesh

12345

68.746002

Structure Pointer

- It is the pointer which points to the address of the memory block that stores a structure.
- **Pointer variable points the address of the variable.**
- Structure pointer points the address of the structure variable.
- **Dot(.) operator is used to access the struct member using normal structure variable**
- **Arrow (->) is used to access the struct member using structure pointer variable.**

Structure Pointer

- Syntax:

```
struct udsn
{
    datatype v1;
    datatype v2;
};
struct udsn sv;
struct udsn *psv;
psv=&sv;
Member Access
psv->v1;
psv->v2;
```

Ex.

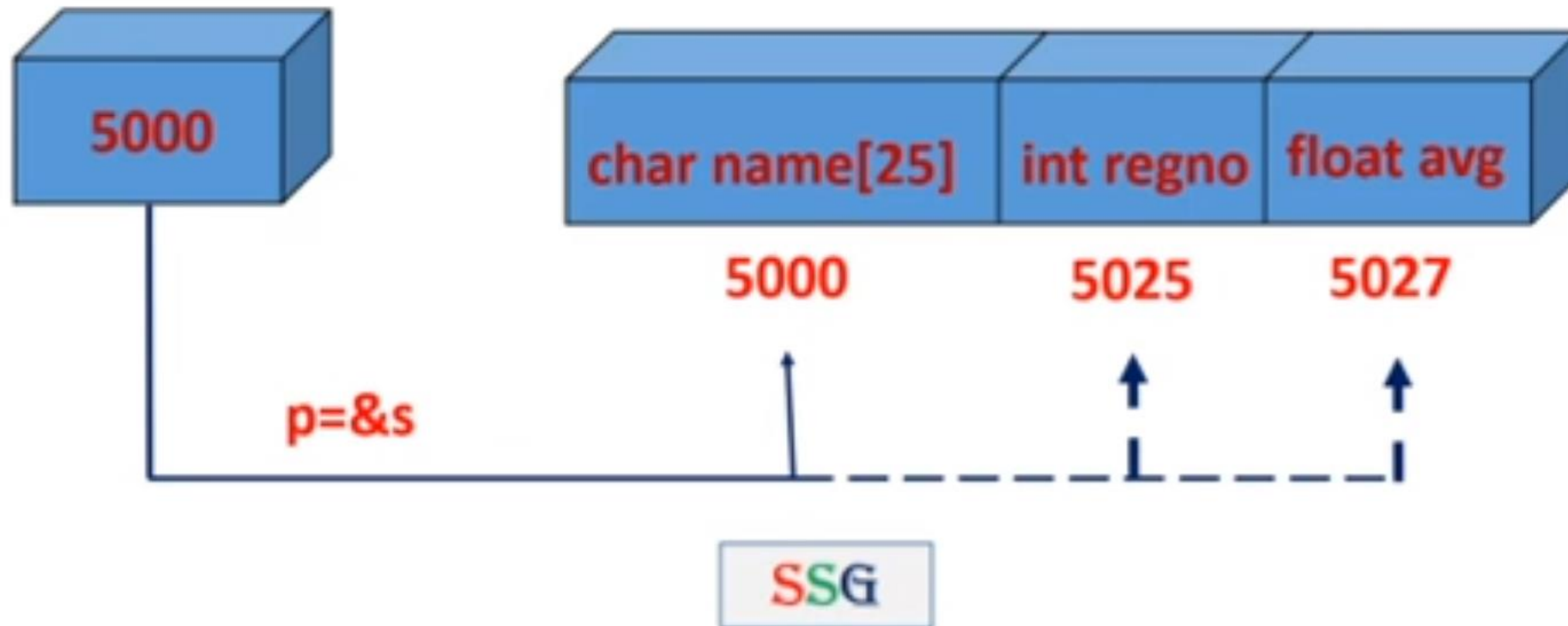
```
struct student
{
    char name[25];
    int regno;
    float avg;
};
struct student s1={"Raja",11223, 65.786};
struct student *p1;
p1=&s1;
Member Access
p1->name, p1->regno, p1->avg;
```

struct student

```
{  
  char name[25];  
  int regno;  
  float avg;  
};
```

Pointer to struct variable
struct student *p;

Structure variable
struct student s;



Program for initialization

```
#include<stdio.h>
struct student
{
    char name[25];
    int regno;
    float avg;
};

void main()
{
    struct student s={"ssg",1122,56.78}, *p;
    p=&s;
    printf("Name\t\tRegNo\t\tAverage\n");
    printf("%s\t\t%d\t\t%f\n",p->name,p->regno,p->avg);
}
```

Name	RegNo	Average
ssg	1122	56.779999

Program for user input

```
#include<stdio.h>
struct student
{
    char name[25];
    int regno;
    float avg;
};
```

```
void main()
{
    struct student s1, *p;
    printf("Enter name, regno and avg\n");
    scanf("%s%d%f",&s1.name,&s1.regno,&s1.avg);
    p=&s1;
    printf("Name\t\tRegNo\t\tAverage\n");
    printf("%s\t\t%d\t\t%f\n",p->name,p->regno,p->avg);
}
```

Enter name, regno and avg

sankarganesh

12345

68.746

Name

RegNo

Average

sankarganesh

12345

68.746002

Program for Separate Structures

```
#include<stdio.h>
```

```
struct date
```

```
{
```

```
    int day, month, year;
```

```
};
```

```
struct student
```

```
{
```

```
    char name[25];
```

```
    int regno;
```

```
    float avg;
```

```
    struct date dob;
```

```
};
```

```
void main()
```

```
{
```

```
    struct student s1,*p;
```

```
    printf("Enter name, regno and avg\n");
```

```
    scanf("%s%d%f",&s1.name,&s1.regno,&s1.avg);
```

```
    printf("Enter day,month and year for dob\n");
```

```
    scanf("%d%d%d",&s1.dob.day,&s1.dob.month,&s1.dob.year);
```

```
    p=&s1;
```

```
    printf("Name\t\tRegNo\t\tAverage\t\tDOB\n");
```

```
    printf("%s\t\t%d\t\t%f\t\t",p->name,p->regno,p->avg );
```

```
    printf("%d-%d-%d\n", p->dob.day, p->dob.month, p->dob.year );
```

```
}
```

Output

Size is = 37

Enter name, regno and avg

sankarganesh

12345

68.746

Enter day,month and year for dob

23

04

1976

Name

RegNo

Average

DOB

sankarganesh

12345

68.746002

23-4-1976

Program for Embedded Structure

```
#include<stdio.h>
struct student
{
    char name[25];
    int regno;
    float avg;
    struct date
    {
        int day, month, year;
    }dob;
};
```

```
void main()
{
    struct student s1,*p;
    printf("Enter name, regno and avg\n");
    scanf("%s%d%f",&s1.name,&s1.regno,&s1.avg);
    printf("Enter day,month and year for dob\n");
    scanf("%d%d%d",&s1.dob.day,&s1.dob.month,&s1.dob.year);
    p=&s1;
    printf("Name\t\tRegNo\t\tAverage\t\tDOB\n");
    printf("%s\t\t%d\t\t%f\t\t",p->name,p->regno,p->avg );
    printf("%d-%d-%d\n", p->dob.day, p-> dob.month, p-> dob.year );
}
```

Outputs

Enter name, regno and avg

sachin

10101

88.9746

Enter day,month and year for dob

23

04

1974

Name

RegNo

Average

DOB

sachin

10101

88.974602

23-4-1974

Program for three struct variables

```
#include<stdio.h>
struct student
{
    char name[25];
    int regno;
    float avg;
};
```

Name	RegNo	Average
sankar	12345	75.750000
ganesh	3456	66.567001
balaji	5678	78.890004

```
void main()
{
    struct student s1={"sankar",12345,75.75};
    struct student s2={"ganesh",3456,66.567};
    struct student s3={"balaji",5678,78.89};
    printf("Name\t\tRegNo\t\tAverage\n");
    printf("%s\t\t%d\t\t%f\n",s1.name,s1.regno,s1.avg);
    printf("%s\t\t%d\t\t%f\n",s2.name,s2.regno,s2.avg);
    printf("%s\t\t%d\t\t%f\n",s3.name,s3.regno,s3.avg);
}
```

Array of Structures

- In the previous program, we have stored data of 3 students in the structure.
- The complexity of the program will be increased if there are 20 students. It means that we have created 20 different struct variables such as s1, s2, s3...s20 and store them one by one.
- An array of structures is an array in which each element is a structure of the same type.
- The referencing and subscripting of these arrays (also called structure arrays) follow the same rules as simple arrays.

Array of Structures

- Syntax:

```
struct udsn  
{  
    datatype v1;  
    datatype v2;  
};  
struct udsn sv[int];
```

- The input, process and output operations are performed using looping statements.

Ex.

```
struct student  
{  
    char name[25];  
    int regno;  
    float avg;  
};  
struct student s[3];
```

- First the structure must be declared and then an array of structure variable should be defined.

Structure Declaration

```
struct student
```

```
{
```

```
char name[25];
```

```
int regno;
```

```
float avg;
```

```
};
```

Array of Structure Variables

```
struct student s[3];
```

S[0]



5001

5026

5028

S[1]



5032

5057

5059

S[2]



5063

5088

5090

Program for user input

```
void main()
{
    struct student s[3];
    int i;
    for(i=0;i<3;i++)
    {
        printf("Enter name, regno and avg of %d student\n",i+1);
        scanf("%s%d",&s[i].name,&s[i].regno);
        scanf("%f",&s[i].avg);
    }
    for(i=0;i<3;i++)
    {
        printf("Address of s[%d] is %u\n",i,&s[i]);
    }
    printf("Name\t\tRegNo\t\tAverage\n");
    for(i=0;i<3;i++)
    {
        printf("%s\t\t%d\t\t%f\n",s[i].name,s[i].regno,s[i].avg);
    }
}
```

```
#include<stdio.h>
struct student
{
    char name[25];
    int regno;
    float avg;
};
```

Outputs

Enter name, regno and avg of 1 student

sankar

1111

65.78

Enter name, regno and avg of 2 student

ganesh

2222

66.77

Enter name, regno and avg of 3 student

balaji

4444

88.67

Address of s[0] is 65432

Address of s[1] is 65463

Address of s[2] is 65494

Name	RegNo	Average
sankar	1111	65.779999
ganesh	2222	66.769997
balaji	4444	88.669998

Program using initialization

```
void main()
{
    struct student s[3]= { {"sankar", 1111, 65.78},
                           {"ganesh", 2222, 66.77},
                           {"balaji", 4444, 88.67}
                           };

    int i;
    for(i=0;i<3;i++)
    {
        printf("Address of s[%d] is %u\n",i+1,&s[i]);
    }
    printf("Name\t\tRegNo\t\tAverage\n");
    for(i=0;i<3;i++)
    {
        printf("%s\t\t%d\t\t%f\n",s[i].name,s[i].regno,s[i].avg);
    }
}
```

```
#include<stdio.h>
struct student
{
    char name[25];
    int regno;
    float avg;
};
```

Outputs

Address of s[0] is 65432

Address of s[1] is 65463

Address of s[2] is 65494

Name	RegNo	Average
sankar	1111	65.779999
ganesh	2222	66.769997
balaji	4444	88.669998